Uchaaran

A little practice. A deeper connection.

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The problem we are solving

Children need help pronouncing Gurbani and a reason to return to daily practice. With Khalis, we focused on pronunciation first, with engagement that sustains practice and understanding that goes beyond repeating words.

Two modes. One path toward mastery.

1. Practice at your pace

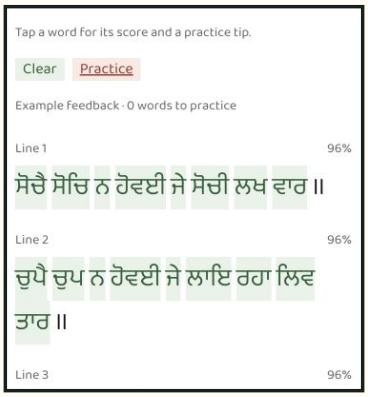
Listen, repeat, and recall with Gurmukhi, Roman-letter Punjabi and English meaning. Focus on a difficult word or practice a whole line.

2. Take the pauri challenge

Recite a line and tap Done. Review word-level feedback, then revisit words that need practice. Stars earn Nanak coins for clothing, belongings and character actions.



Listen + recall



Find the words to practice next.

Focused feedback supports the next attempt.

Review each word

What changed since midterm

Organizers asked for less pressure and simpler feedback. We added line-by-line scoring, short word prompts, and an ML maintenance guide for Khalis.

Why a memory palace?

For grades 6 to 8, familiar scenes link each line to a place along 38 pauris. Sikh-inspired journeys invite recall and reflection. The ustaad guides pronunciation; rewards follow learning, not screen time.

Tools and implementation

Expo · React Native · TypeScript · Azure-hosted IndicConformer · BaniDB · reference audio. Codex and Paper supported development and design.

Private processing, by design

Recordings are processed privately in Azure. Temporary server recordings are deleted after scoring, with no use for public-model training.

Khalis can host the model in its own shared AWS or Azure environment. On-device packaging is another option for fully local processing that keeps recordings on device.

Ready for handover. MIT source, setup and ML maintenance docs included. Khalis supplies hosting and credentials; costs depend on usage. An ongoing technical owner remains to be named.

Watch demo ↗ | **Try the app ↗** | **View code ↗** | **Design process ↗**